

Assignment Title:

Time Series Modelling Case Study

**Forecasting German Electricity Demand Using
Statistical, Machine Learning and Deep Learning
Models**

Module Name

Time Series Analysis

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GitHub Link: <https://github.com/PatelSandeep9515/Time-Series-Modelling-Case-Study>

Abstract

Proper prediction of electricity consumption is critical in the effective management of power systems. In this paper, we compare statistical, machine learning, and deep learning models in predicting German electricity consumption with Open Power System Data (2015-2020). Weekly data was used for benchmarking methods including SARIMA, SARIMAX, Random Forest, and Gradient Boosting, whereas hourly data was used for training an LSTM model. Evaluation was done using Mean Absolute Error, Root Mean Square Error, Mean Absolute Percentage Error, and coefficient of determination (R^2). The results indicate that machine learning algorithms outperform the use of statistical models, where the LSTM model achieves the best results (RMSE = 960.69, R^2 = 0.9907). The results prove the effectiveness of advanced learning algorithms in capturing nonlinearity and temporal characteristics in the prediction of electrical power consumption.

Keywords: Electricity Consumption Prediction, Time Series Analysis, SARIMA, SARIMAX, Random Forest, Gradient Boosting, LSTM.

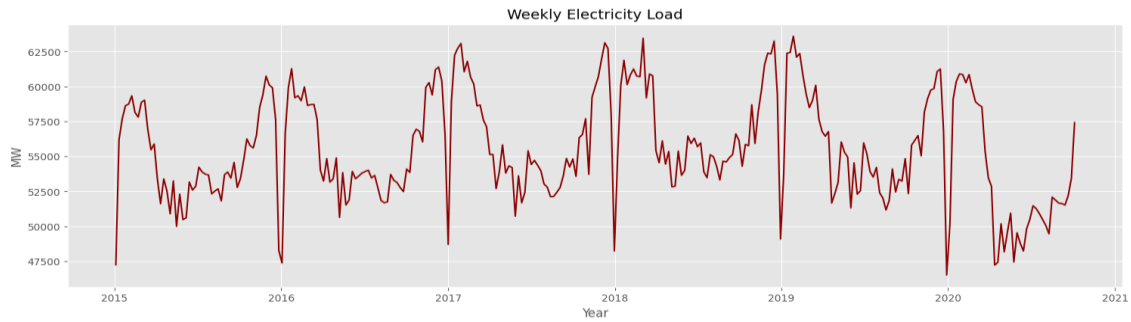
1. Introduction

Forecasting of electrical demand plays an important role in planning of electric power system, generation schedule and functioning of the electricity market. Since electricity demand depends on various factors like weather conditions, seasonality and consumer behaviour, better forecasting will lead to higher reliability of electricity system and its better operation. While statistical methods like SARIMA are able to account for trends and seasonality, they cannot deal well with the nonlinearity in demand. Machine learning and deep learning algorithms are able to capture complicated relations through training on historical data; thus, they are appropriate for electricity demand forecasting. The research compares different approaches: benchmark methods, SARIMA, SARIMAX, Random Forest, Gradient Boosting and Long Short-Term Memory (LSTM) through analysing of electricity demand in Germany. Weekly data was applied for statistical and machine learning algorithms, while hourly data – for LSTM.

2. Dataset and Exploratory Data Analysis

Data related to the electricity demand of Germany, spanning between January 2015 and October 2020, has been considered in the OPSD dataset. In the case of the benchmark models, namely SARIMA, SARIMAX, Random Forest, and Gradient Boosting Models, the data were first pre-processed by converting time stamps, handling any missing values, and aggregating on weekly average basis. However, in LSTM, hourly data were used.

Figure 1. Weekly German Electricity Demand



It can be seen from Figure 1 that there is a distinct seasonal trend in a year, where higher consumption is recorded in winter than in the warm months. Short-term variations indicate the presence of some external influences such as climate and public holidays. Such features explain why seasonal and sophisticated forecasting techniques should be applied.

Stationarity Analysis

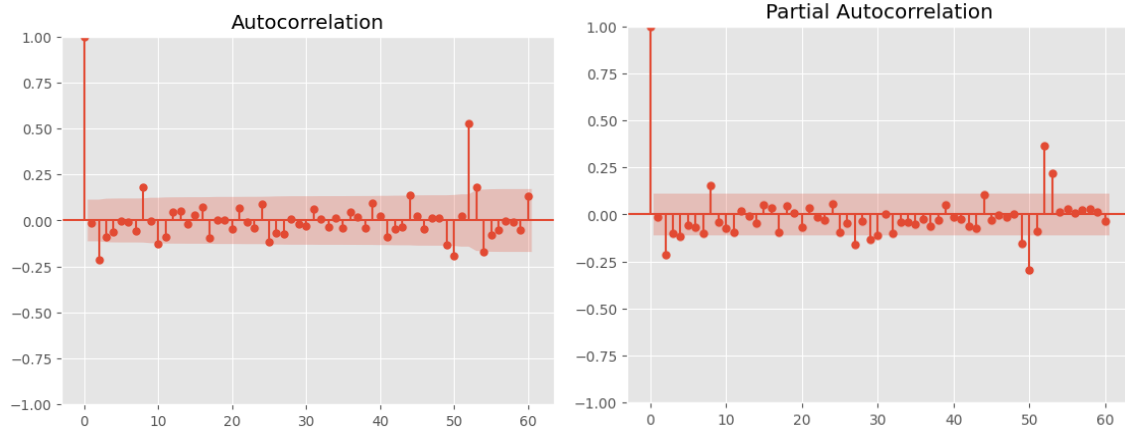
Since autoregressive forecasting models are dependent on the assumption of stationarity, the electricity demand time series was analysed using the Augmented Dickey–Fuller (ADF) test alongside autocorrelation tests. Stationarity tests help to determine whether the time series data have constant statistical properties over time, which is a key prerequisite of SARIMA modelling.

Table 1. Augmented Dickey–Fuller Test Results

Statistic	Value
ADF Statistic	-7.0693
p-value	4.9847×10^{-10}
Critical Value (1%)	-3.4529
Critical Value (5%)	-2.8715
Critical Value (10%)	-2.5721

ADF test yielded a p-value well below the 0.05 level of significance, showing that the electricity demand data when differenced is stationary. This means that the null hypothesis of presence of a unit root was rejected, thus meeting the conditions necessary for the autoregression model.

Figure 2. Autocorrelation Function (ACF) & Figure 3. Partial Autocorrelation Function (PACF)



ACF plot shows high levels of autocorrelation at various lags, which means that there is an indication that current level of demand for electricity is highly affected by previous values. PACF plot also shows peaks during early lags, which suggests the existence of autoregressive behaviour. Both ADF, ACF and PACF plots justify the use of seasonality based autoregressive models to predict future demand for electricity.

3. Methodology

The statistical, machine learning, and deep learning models were trained using a chronological train-test split where the last two years were left out for testing. The weekly dataset was used to train the benchmark, SARIMA, SARIMAX, Random Forest, and Gradient Boosting models whereas the hourly dataset was used to train the LSTM model.

3.1 Forecasting Benchmark Models

The four benchmark models (mean, naïve, seasonal naïve, and drift) have been created to determine a basis of comparison against which the performance of more sophisticated forecasting methods would be tested.

3.2 SARIMA Model

The SARIMA model was built to consider the trend, seasonality, and temporal dependence of data. SARIMA parameters were chosen based on the AIC grid search following ADF, ACF and PACF tests of stationarity.

3.3 SARIMAX Model

SARIMAX is an extension of the SARIMA model with exogenous data – the weekly temperature in Berlin. The temperature values were known beforehand in the test phase.

3.4 Feature Based Machine Learning Models

The Random Forest and Gradient Boosting algorithms were built using lagged demand, calendar variables, and weekly temperatures. Hyperparameter tuning was done to optimize forecasting performance and avoid overfitting.

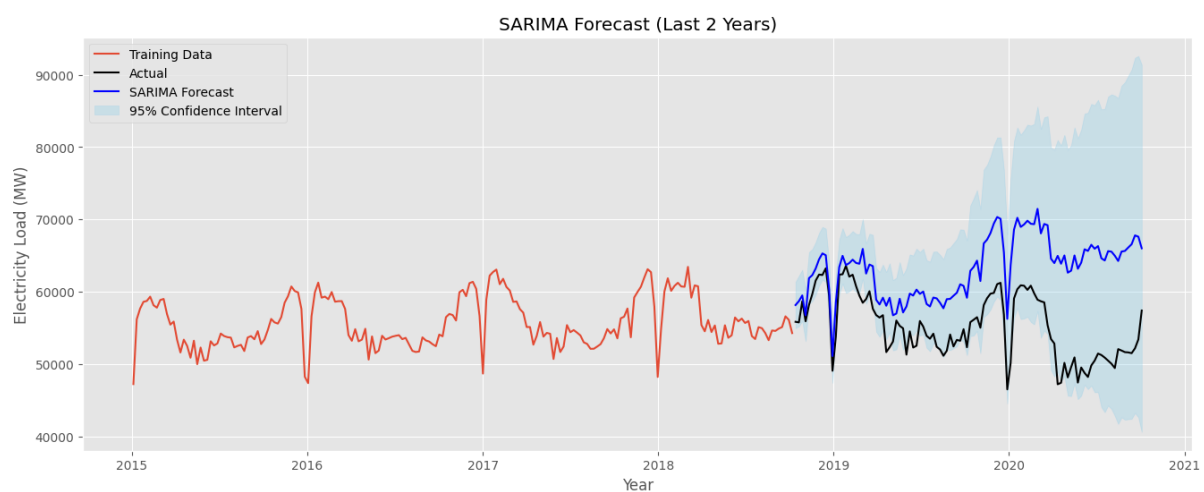
3.5 Long Short-Term Memory (LSTM)

The LSTM algorithm was implemented based on the hourly electricity demand dataset. It uses normalized sequences and early stopping to take advantage of its capability in learning temporal relations.

4. Results & Critical Discussion

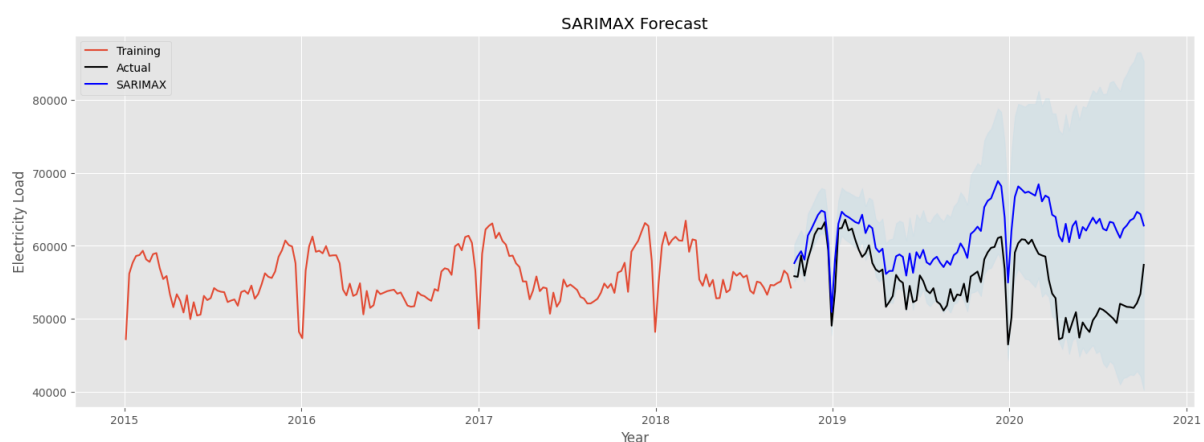
Forecasting ability was compared among the models in the last two years of data collection. Baseline forecasting accuracy was set with the benchmark models prior to comparison with the statistical, machine learning, and deep learning methods. Forecast accuracy was measured through MAE, RMSE, MAPE, and R^2 .

Figure 4. SARIMA Forecast



SARIMA captured the seasonality trend but generated larger forecast error values, implying that linear autoregressive models cannot fully capture the non-linear nature of electricity demand.

Figure 5. SARIMAX Forecast

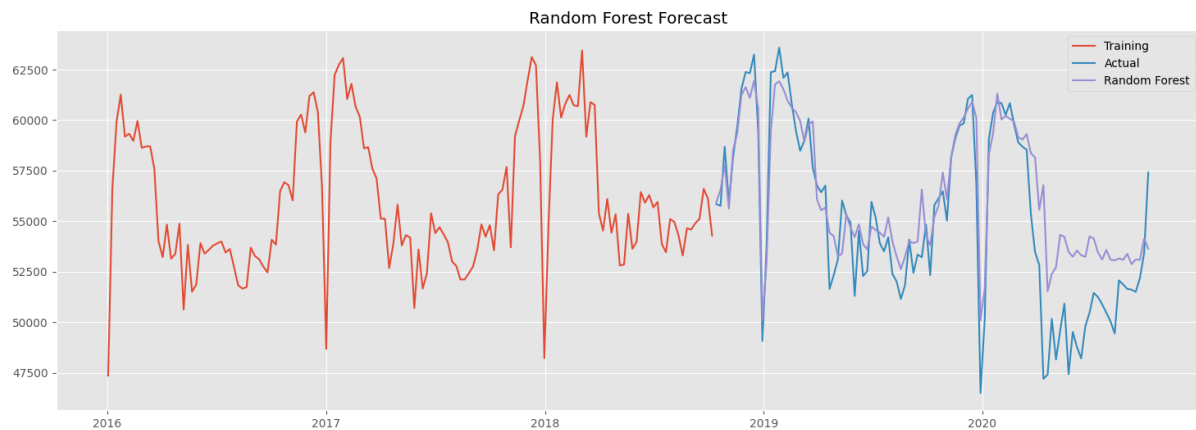


The inclusion of temperature helped improve forecast accuracy, showing that weather is valuable information for prediction. However, the results still fell short of the performance of machine learning algorithms.

4.1 Machine Learning Algorithms Based on Features

Models of Random Forest and Gradient Boosting algorithms have been developed using engineered time series features and temperature per week to model nonlinear relations of electricity demand.

Figure 6. Random Forest Forecast

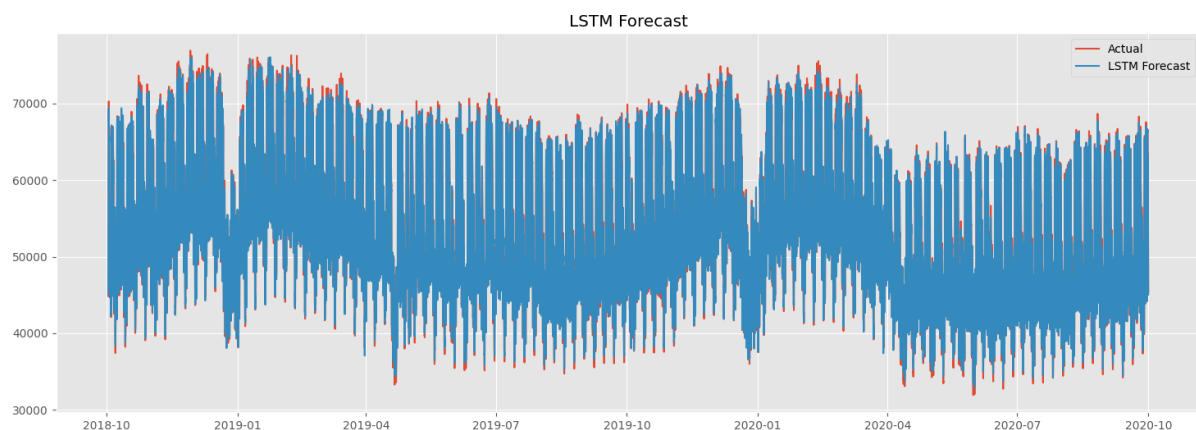


The Random Forest algorithm managed to model the seasonal electricity demand very well and surpassed the performance of the statistical models due to its ability to model nonlinear relations. The Gradient Boosting algorithm performed better in terms of accuracy with smaller values of MAE and RMSE. This shows how effective the ensemble learning is in electricity demand prediction. Both of these algorithms are very dependent on good feature engineering.

4.2 Long Short-Term Memory (LSTM)

In contrast to the statistical models, the LSTM network was trained on the raw data (hourly electricity demand data).

Figure 7. LSTM Forecast



The LSTM had the highest accuracy in forecasting, providing predictions very close to the observed values. The model managed to capture long-term dependencies in hourly data and, consequently, to perform better compared to all other models.

4.3 Forecast Comparison

The performance of each of the models was estimated based on four different performance criteria: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2).

Table 2. Forecast Evaluation Metrics

Model	MAE	RMSE	MAPE	R^2
Mean	3788.83	4397.30	0.0697	-0.0121
Naïve	3783.20	4459.11	0.0679	-0.0408
Seasonal Naïve	2318.52	3006.76	0.0441	0.5268
Drift	4339.89	5117.96	0.0805	-0.3710
SARIMA	8077.82	9434.25	0.1519	-3.6588
SARIMAX	6738.31	7825.29	0.1268	-2.2052
Random Forest	1619.51	2258.79	0.0310	0.7329
Gradient Boosting	1600.58	2200.89	0.0306	0.7465
LSTM	731.57	960.69	0.0135	0.9907

Seasonal naïve showed the best results for the statistical models, confirming the presence of seasonality in German electricity demand. While SARIMA was able to account for the presence of seasonality, its assumptions about linearity caused high prediction errors. Introducing temperature information improved the performance of SARIMAX, however, it still did not perform as well as machine learning approaches. The Random Forest and Gradient Boosting models were able to account for nonlinear relations in the data. The best results among all the models were achieved by LSTM (RMSE = 960.69, R^2 = 0.9907).

Figure 8. Comparison of Forecasts

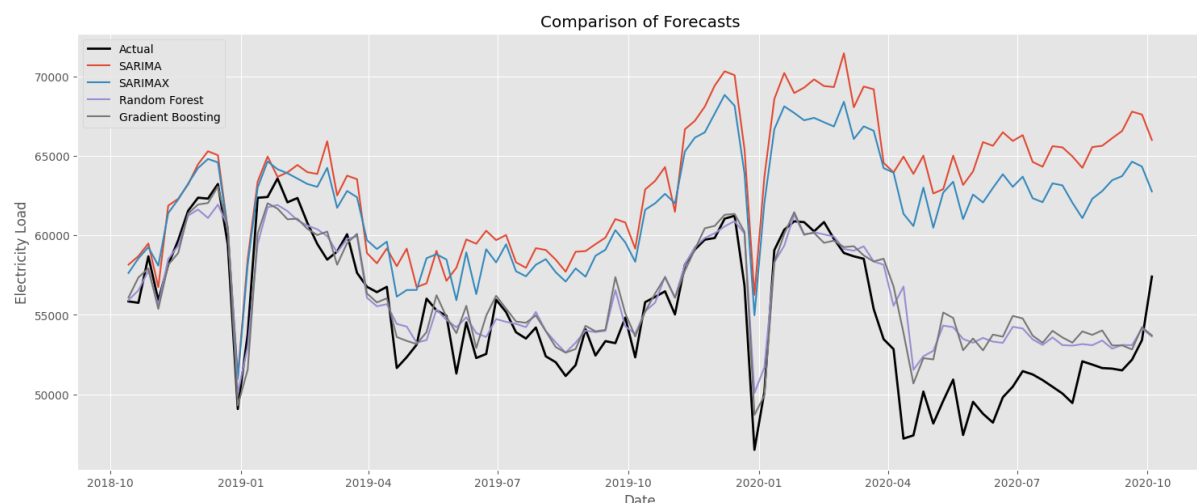


Figure 8 is a comparison of the different forecasting techniques used in the testing phase. The benchmark models offer some level of forecast but cannot incorporate the details of the temporal behaviour. SARIMA and SARIMAX follow the same trend as seen but have large variations when there are fluctuations in the demand for electricity. The prediction by the

Random Forest and Gradient Boosting technique follow the demand seen quite closely whereas the LSTM model predictions overlap almost all of the actual values in the test phase.

5. Conclusion

This study considered different approaches of statistical models, machine learning models, and deep learning models for forecasting of the German electricity demand. Although Seasonal Naïve proved to be the best benchmark, machine learning models outperformed these models in terms of forecasting accuracy. LSTM model showed the lowest forecasting error (RMSE = 960.69; $R^2 = 0.9907$). Although SARIMA and SARIMAX models have an advantage in interpretability, their poor performance is due to the non-linear nature of electricity demand. Thus, LSTM model can be used for operational forecasting, and Gradient Boosting as a good trade-off.

6. Future Work

Future work may consider incorporating more features, such as holidays, electricity prices, and renewable energy into forecasting model as well as other architectures, like GRU and Transformer networks.

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